



AFRL-AFOSR-VA-TR-2026-0080

Effect of current sheath initiation on the radial collapse and energetic particle acceleration in 10 kJ Dense Plasma Focus

Farhat Beg
UNIVERSITY OF CALIFORNIA, SAN DIEGO
9500 GILMAN DR
LA JOLLA, CA, 92093
USA

12/15/2025
Final Technical Report

DISTRIBUTION A: Distribution approved for public release.

Air Force Research Laboratory
Air Force Office of Scientific Research
Arlington, Virginia 22203
Air Force Materiel Command

REPORT DOCUMENTATION PAGE

PLEASE DO NOT RETURN YOUR FORM TO THE ABOVE ORGANIZATION.

1. REPORT DATE 20251215		2. REPORT TYPE Final		3. DATES COVERED	
				START DATE 20190201	END DATE 20220731
4. TITLE AND SUBTITLE Effect of current sheath initiation on the radial collapse and energetic particle acceleration in 10 kJ Dense Plasma Focus					
5a. CONTRACT NUMBER		5b. GRANT NUMBER FA9550-19-1-0073		5c. PROGRAM ELEMENT NUMBER 61102F	
5d. PROJECT NUMBER		5e. TASK NUMBER		5f. WORK UNIT NUMBER	
6. AUTHOR(S) Farhat Beg					
7. PERFORMING ORGANIZATION NAME(S) AND ADDRESS(ES) UNIVERSITY OF CALIFORNIA, SAN DIEGO 9500 GILMAN DR LA JOLLA, CA 92093 USA					8. PERFORMING ORGANIZATION REPORT NUMBER
9. SPONSORING/MONITORING AGENCY NAME(S) AND ADDRESS(ES) Air Force Office of Scientific Research 875 N. Randolph St. Room 3112 Arlington, VA 22203				10. SPONSOR/MONITOR'S ACRONYM(S) AFRL/AFOSR RTB1	11. SPONSOR/MONITOR'S REPORT NUMBER(S) AFRL-AFOSR-VA-TR-2026-0080
12. DISTRIBUTION/AVAILABILITY STATEMENT A Distribution Unlimited: PB Public Release					
13. SUPPLEMENTARY NOTES					
14. ABSTRACT Final report for grant that ended in FY22.					
15. SUBJECT TERMS					
16. SECURITY CLASSIFICATION OF:				17. LIMITATION OF ABSTRACT UU	18. NUMBER OF PAGES 20
a. REPORT U	b. ABSTRACT U	c. THIS PAGE U			
19a. NAME OF RESPONSIBLE PERSON JOEL BIXLER					19b. PHONE NUMBER (Include area code) 00000000

REPORT COVER SHEET

Award Number: FA9550-19-1-0073

Report Type: Final Report

Reporting Period: 02/01/2019 – 01/31/2021

Distribution Statement: Distribution A – Approved for Public Release

Program Officer: Dr. John Luginsland, US Air Force Office of Scientific Research

Principal Investigator: Dr. Farhat Beg, Center for Energy Research, University of California San Diego

Project Title: Effect of Current Sheath Initiation on the Radial Collapse and Energetic Particle Acceleration in 10 kJ Dense Plasma Focus

Abstract: The Dense Plasma Focus (DPF) is a pulsed power-driven plasma discharge that emits X-rays and energetic particles (electrons, ions and neutrons). The pinch dynamics involves a surface breakdown at the insulator sleeve that separates the anode and cathode in a coaxial geometry. Then, the plasma peels off of the insulator surface due to the Lorentz ($\mathbf{J} \times \mathbf{B}$) force and moves along the coaxial electrodes, sweeping the background gas with it. At the end of this run-down phase, when the plasma sheath reaches the end of the electrodes, it turns over and compresses on axis. Charged particles in these devices are accelerated by intense electromagnetic fields produced by instabilities formed during the final radial compression phase. Even though instability formation and particle acceleration are well-known processes in DPFs, open questions regarding pinch dynamics and particle generation remain, warranting further investigation. Here, we report on the experimental and numerical work on the role of dopant gases, insulator sleeve length, and anode geometries on X-ray and neutron generation. The impact of the work is that it enabled a better understanding of the underlying physics of pinch dynamics and neutron generation, providing validation of models and scaling laws. In addition, funding facilitated training of young workforce in High Energy Density Science.

1. Executive Summary

The Dense Plasma Focus (DPF) is a pulsed power-driven plasma discharge that emits copious X-rays and energetic particles, such as electrons, ions and neutrons. The discharge is initialized from a surface breakdown at the insulator sleeve that separates the anode and cathode in a coaxial geometry. Then, the plasma peels off, of the insulator surface due to the Lorentz ($\mathbf{J} \times \mathbf{B}$) force and moves along the coaxial electrodes, sweeping the background gas with it. At the end of this run-down phase, when the plasma sheath reaches the end of the electrodes, it turns over and compresses on axis. Charged particles in these devices are accelerated by intense electromagnetic fields produced by instabilities formed during the final radial compression phase. Neutrons, on the other hand, are mainly generated by the beam-target mechanism as a byproduct of the instabilities. Even though instability formation and particle acceleration are well-known processes in DPFs, open questions regarding particle generation remain, warranting further investigation.

The gas pressure, dopant gas and its percentage and anode length can significantly affect the quality of the pinch and subsequently neutron yield. There has not been a comprehensive effort to systematically study the underlying physics of above-mentioned effects on the pinch dynamics and neutron generation. Experimental and modeling work is therefore warranted to address the critical aspect of pinch dynamics.

In this report, we describe experimental and numerical work on the role of dopant gases, insulator sleeve length and anode geometries on X-ray and neutron generation. We carried out experiments to study the role of dopant gas (krypton) at the 265-346 kJ Gemini DPF (Nevada National Security Site). Experimental time of the Gemini DPF is restricted, and it is not open to the outside users. However, we were granted experimental time due to our ideas on enhancement of neutron yield using dopant gases. Indeed, our experiments showed that a dopant gas not only affects pinch dynamics but also improves neutron yield. We showed that the optimal concentration of Kr at the MA level is near 0.1% by volume—elucidating a trend in the optimal Kr doping concentration as a function of the device scale. The neutron time-of-flight data revealed that Kr doping creates shorter and more intense neutron bursts, likely due a tighter but unstable pinch, highlighting a key trade-off for Kr doping. These results are consistent with MHD modeling that showed instabilities are more inherently 3D in nature and/or seeded by kinetic effects.

Experiments to investigate the role of insulator sleeve length and anode geometries were carried out using a newly constructed 4.6 kJ, 20 kV, 230 kA DPF at UC San Diego. We found that the neutron yield is significantly dependent on the insulator length. As the length of the insulator increases, the exposed anode length traveled by the sheath during the run-down phase decreases. This suggests an increase in the optimal fill pressure with increasing insulator length to maintain the pinch time near peak current. However, in our work, the opposite trend was observed—the optimal fill pressure for neutron production decreased with increasing insulator length. Optical probing of the sheath from run-down to the pinch reveals significant plasma filamentation with increasing pressure and a dependence of insulator length on filamentation onset. For the anode geometries study, we used solid and hollow anodes with either a flat or hemispherical tip. We showed for the first time evidence of the extension of the pinch column inside the hollow anode, presumably responsible for higher neutron yield. A maximum neutron yield of $(2.0 \pm 0.2) \times 10^8$ neutrons per pulse was observed with the hollow hemispherical anode, while higher hard X-ray emissions were observed for the solid hemispherical anode. This analysis showed that a hollow anode optimizes neutron yield while a solid anode serves as an enhanced X-ray source due to electron beams.

The impact of the work is that it enabled a better understanding of the underlying physics of the pinch dynamics and neutron generation, providing validation of models and a discovery of a new scaling law. This work facilitated training of graduate students (David Housley, Veronica Eudave, Jackie Vaughan, and Swarvanu Ghosh) and post-doctoral fellow (Jeff Narlis and Eric Hahn) in MHD modeling, pulsed power-driven experiments and plasma diagnostics.

2. Introduction

High Energy Density Plasmas can be produced by powerful lasers, particle beams, or Z-pinchs. One of the characteristic properties of Z-pinchs is high conversion efficiency of stored electrical energy into the plasma. The high intrinsic efficiency of dense Z-pinchs has been extensively used for generation of soft X-ray radiation [1-2]. However, Z-pinchs could also be very efficient sources of energetic electrons, ions and neutrons. In particular, there is a growing interest in pulsed sources of ions and neutrons. Small, repetitive, and portable neutron sources could be useful in radiation material science, radiobiology, nuclear medicine, cargo inspection, and illicit material detection, whereas large devices could be effective for the recycling of nuclear waste and for a fusion-fission hybrid reactor, etc. The advantage of employing neutrons is that they can penetrate through chamber walls, whereas troublesome debris produced by a Z-pinch remains within an experimental chamber. Thus, it seems natural to conduct research on Z-pinchs, not only as X-rays sources, but also as efficient plasma-based ion and neutron sources. To obtain sufficient number of ions and neutrons for these applications, various configurations based on the Z-pinch effect have been suggested and tested. The most promising configuration appeared to be the Dense Plasma Focus (DPF), operated with deuterium gas.

The DPF is a simple pulsed power device capable of producing intense pulses of radiation and energetic charged particles. These devices are also an excellent source of neutrons as a product of D-D nuclear reactions when deuterium gas is used. The DPF was developed by Filippov [3] in 1962 and by J. W. Mather [4] in 1964 while doing work with coaxial plasma accelerators. Although both designs are very similar, the main difference is the ratio of the electrodes' length to radius (>1 for the Mather type and less than 1 for Filippov's). Most of the work on DPF has been done on the Mather type and henceforth, we will be focusing on this geometry.

The DPF uses a capacitor bank that is discharged in open-ended coaxial electrodes (Fig. 1). When the switch closes, breakdown takes place at the insulator sleeve and a current sheath forms at the base of the electrodes (Fig. 1a). The $\mathbf{J} \times \mathbf{B}$ force drives the current sheath upward, snowplowing the gas in front of it as it moves along the electrodes (Fig. 1b). When the current sheath reaches the top of the anode, it radially collapses (Fig. 1c), producing hot (0.5-2 keV) and dense ($\sim 10^{19} \text{ cm}^{-3}$) plasma. Besides X-ray emission, charged particles (and possibly neutrons) are produced.

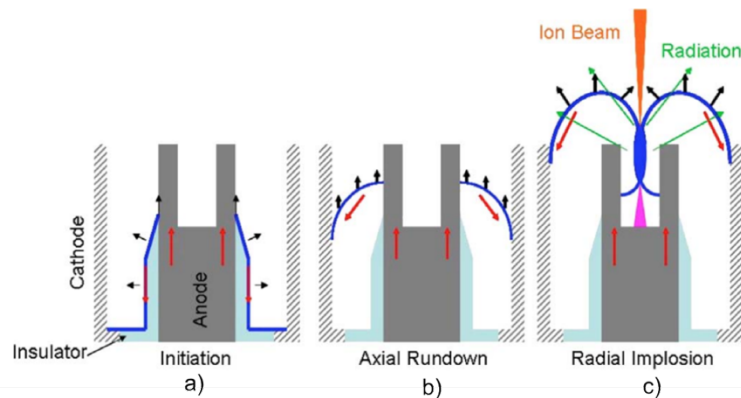


Figure 1: Schematic of the Dense Plasma Focus device. The schematic shows breakdown (a), axial (b) and radial collapse (c) phases. Pictures taken from ref. [1].

Fig. 2 shows a typical time sequence of the final radial motion of the plasma sheath in a UC San Diego DPF configuration. Figure displays the sheath propagating radially inwards to form a dense pinch, which becomes MHD unstable later on and finally, the formation of an axial shock wave propagating into the cold gas.

As mentioned above, particle emission in DPF is a familiar phenomenon, but at present, there is no consensus on what the acceleration mechanisms are.

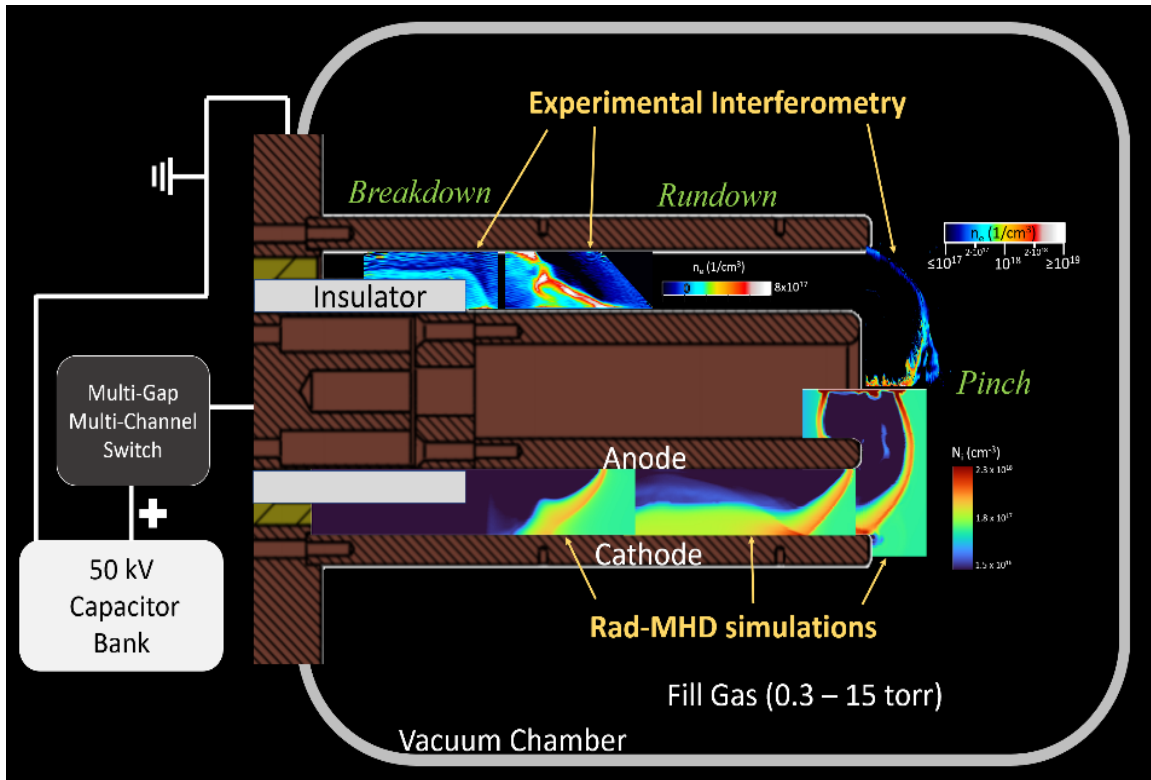


Figure 2: Schematic of the UC San Diego dense plasma focus device showing breakdown, axial rundown, and radial collapse/pinch phases. Experimental images from commissioning studies (top) and benchmarked rad-MHD simulations (bottom) are included in the figure.

One of the remarkable features observed in DPFs is that the ion (and electron) spectrum is not thermal in nature and that the particles' energy is well above the initial voltage of the capacitor bank. The ratio of the ion and electron density N_i/N_e is observed to be compatible with the predicted value for a uniform-current density diode accelerator [6], suggesting the idea that electric fields, likely produced by necking, or $m=0$, instabilities, are responsible for particle acceleration. Necking instabilities are always seen in the final pinch implosion (Fig. 2 c-d). An electron beam peak current of ~ 20 kA has been measured in a ~ 10 kJ DPF with beam pulse width of ~ 50 ns, and a power law energy distribution of ~ -3.8 [7].

The main two topics in DPF that require further study are: i) the role of instabilities on particle acceleration and subsequently neutron generation and ii) role of insulator length and iii) anode geometries on breakdown phase and neutron generation. We addressed these topics using two different devices: MSTs GEMINI (265–346 kJ, 2 MA, 6 μ s) and UCSD DPF (10 kJ, 230 kA, 2 μ s). Experimental time of the Gemini DPF is restricted, and it is not open to the outside users. However, we were granted experimental time due to our ideas on enhancement of neutron yield using dopant gases.

3. Progress Report

We extensively studied experimentally and numerically the role of impurities (dopant gases) on instabilities and neutron beam generation. In DPF, instabilities are responsible for generating electron and ion beams. The ion beam generates non thermal neutrons. We also studied the role of anode geometries on X-ray and neutron production.

Role of dopant gases in deuterium for neutron production:

Numerous studies on drivers of widely varying energies have found that doping a deuterium-fill DPF with Ne, Ar, or Kr affects the hydrodynamic evolution of the plasma sheath and/or the yield/pulse shape of the neutron signal. The conventional interpretation of these collective results is that doping a deuterium-fill DPF with a fraction (1%–10%) of a high-Z element will increase radiative cooling and lead to a tighter pinch and higher neutron yield, particularly in larger devices where radiative losses can play a larger role.

We investigated the effect of Kr dopant on sheath dynamics and neutron yield in simulations of a 2–3 MA dense plasma focus with 2D radiation-MHD code. Consistent with the conventional understanding, addition of highly radiative Kr dopant increased radiative cooling and resulted in a narrower sheath throughout the axial run-down phase and during the radial implosion phase. Despite the narrowing of the sheath, particularly with 1% Kr as compared to 0.1% Kr, quantifiable instability growth was not observed prior to the pinch, which is consistent with the “self-healing” properties of the DPF, i.e., the tendency for significant non-uniformities to be smoothed out rather than grow in time.

This could also indicate that instabilities are more inherently 3D in nature and/or seeded by kinetic effects. The difference between sheath ion and electron temperature is shown to increase with dopant concentration due to greater radiative cooling, and to decrease with increasing charging voltage due to electron–ion equilibration, though, for both 0.1% Kr and 1% Kr, the difference between T_i and T_e persists throughout the radial implosion phase.

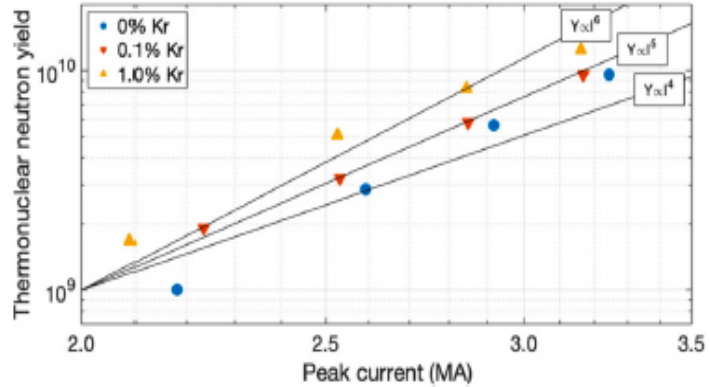


Figure 3: Thermonuclear neutron yield for the 0% Kr (blue), 0.1% Kr (red), and 1.0% Kr (yellow) simulations as a function of peak current. Each cluster of three data points correspond to one of the charging voltages: 35, 40, 45, or 50 kV. Also shown in black are reference lines of yield scaling as a I^4 ; a I^5 , and a I^6 with a reference yield of 10^9 at 2 MA. Note that experimental yields at the 2 MA level, which include beam-target neutrons, are of order 10^{11} .

Fig. 3 shows increase in neutron yield with the increase in Kr dopant concentration at each of the current levels. Neutron yields for the 35 kV simulations with 0% Kr and 1.0% Kr are lower than expected, assuming conventional I^4 scaling. Including these points, power law fits for the 0% Kr, 0.1% Kr, and 1.0% Kr cases are 5.7, 4.64, and 4.85, respectively. The single exponent for all three sets that minimizes total variance is 5.10. Excluding the 35 kV simulations gives values of 5.40, 4.85, and 4.01 for the single-concentration exponents of the 0%, 0.1%, and 1.0% cases, respectively, and 4.75 for the overall minimal-variance exponent. These exponents are perhaps greater than expected, but the overall magnitudes (10^9 – 10^{10}) are consistent with fully kinetic simulations that predict thermonuclear neutron production is a small (<0.01) fraction of the total (10^{11} DD, 10^{12} DT) neutrons produced at similar current levels.

The neutrons in these 2D MHD simulations are produced over a few ns, as compared to the timescales of tens or hundreds of ns in kinetic simulations and experiments, which can be attributed to a combination of a longer pinch duration (possibly due to the absence of 3D instabilities) and the broader beam-target energy spectrum relative to the narrow thermonuclear spectrum.

We used the MSTS GEMINI DPF as shown in Fig. 4 to experimentally test the effect of dopants. The DPF electrodes have diameters 6 in. and 8 in. for the anode and cathode, respectively. Additionally, the anode features a truncated hemispherical top. The cathode is a continuous cylinder, increasing mass accumulation by mitigating losses that can be as much as 41% during the rundown phase. The MSTS system on which

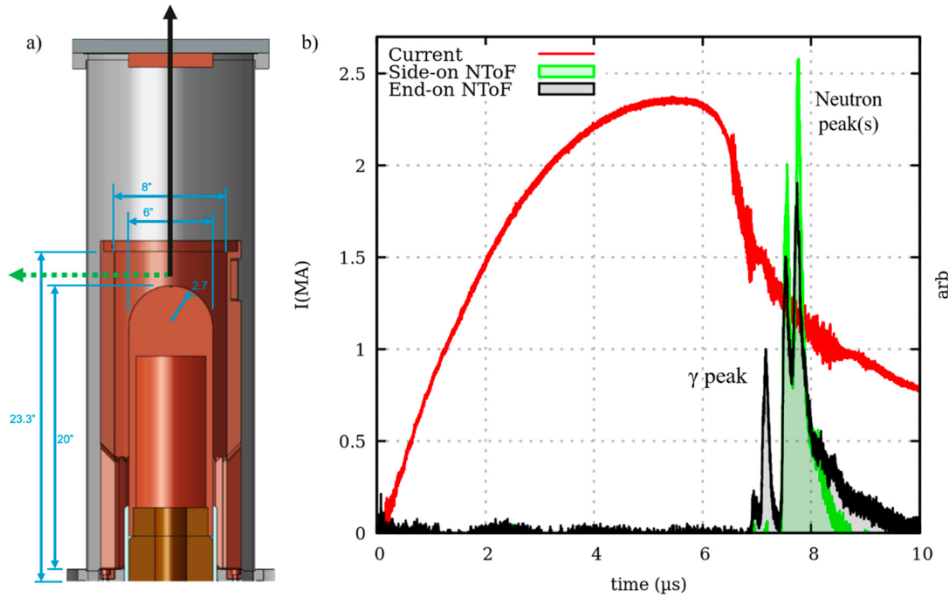


Figure 4:(a) DPF chamber fielded for the present experiments. The anode (center electrode with truncated hemispherical head) and the cathode (cylindrical outer electrode tube) are made of copper, the insulator (light blue) is a machinable quartz, and the vacuum chamber is 0.125 in. stainless steel (SS). Neutrons leaving the pinch in the end-on direction also pass through a 1 in. Cu beam stop and 1 in. SS endplate. Neutron time-of-flight diagnostic paths are shown as the dashed green (side-on/radial), and solid black (end-on/axial) lines in (a) with a representative current (red line) and NToF traces (green and black for side and end-on, respectively) during operation at 2.29 MA for a pure deuterium shot at 35 kV and 9 Torr shown in (b).

these experiments were conducted uses a 432 μF capacitor bank storing 265–346 kJ at a 35–40 kV charge voltage. The drive capacitor bank inductance is ~ 40 nH. With a plasma load, the peak current is typically greater than 2MA and occurs ~ 6 μs after triggering the capacitor bank switches. The capacitor bank charge voltage is measured using a DC high voltage probe, and the current is measured near the vacuum tube header with a calibrated Pearson current monitor as well as a Rogowski coil. A representative current trace (integrated Rogowski) and neutron time-of-flight (NToF) signal are shown in Fig. 4(b). The device was operated over a range of fill pressures (~ 5 –10 Torr) and Kr admixture percentages (0%–1%) at two nominal charging voltages, 35 kV and 40 kV. Both neutron activation and neutron time-of-flight detectors were fielded. The total yield for each shot was measured using Be and Ag activation counters. The NToF measurements are taken both radially and axially at a distance of 12 m from the pinch. The end-on and side-on measurements are attenuated by 1 in. Cu + 1 in. stainless steel and 1 in. Cu + 1/8 in. stainless steel due to their respective lines of sight. To correct this, the raw NToF measurements are multiplied by 2.50 (end-on) and 1.55 (side-on) according to the Beer–Lambert relation for the effective neutron flux through dense matter.

In these experiments, we used 35 kV and 40 kV charging voltages with pure D_2 gas and D_2 with added Kr dopant at varying volume percentages. At 35 kV and 40 kV, the corresponding peak currents are 2.29 ± 0.07 MA and 2.66 ± 0.03 MA with minimal significant difference observed as a function of fill pressure or dopant additions. To maintain consistency in timing, rundown times were managed between 6 and $6.8 \mu s$ [Fig. 5(a)] with a few shots outside of this range taken in order to explore potential timing effects on the yield. The measured neutron yields for the explored ratios of the D_2 –Kr admixture at 2.29 MA are shown in Fig. 5(b). A minimal enhancement of the neutron yield is identified at 0.1% Kr additions with no discernable trend in the yield with a rundown time (controlled via fill pressure in the tube) between 5.7 and $6.8 \mu s$. The average yields for pure deuterium, 0.05%, 0.1%, and 1% Kr additions for rundown times of $\sim 6.4 \pm 0.32 \mu s$ are 2.39×10^{11} , 2.43×10^{11} , 2.68×10^{11} , and 1.15×10^{11} per pulse. Shots higher than 0.5% Kr showed a significant decrease in the yield as compared to the pure deuterium shots [Fig. 5(a)]. Shots taken with pure D_2 at 2.29 MA and 2.66 MA also provide a current scaling relationship consistent with, but slightly less than, I^4 scaling, measured to scale with the current to the 3.49 ± 0.37 power [dashed blue line in Fig. 5(b)]. Unlike at 2.29 MA, moderate Kr additions increase the yield up to $\sim 2 \times$ when operating at 2.66 MA, with a maximum neutron yield measured of 6.9×10^{11} per pulse for 0.1% Kr at 11 Torr with a rundown time of $6.56 \mu s$. At 2.66 MA, the average yields for pure deuterium, 0.05%, 0.1%, and 1% Kr additions for rundown times of $\sim 6.4 \pm 0.32 \mu s$ are 3.34×10^{11} , 4.44×10^{11} , 5.68×10^{11} , and 5.15×10^{11} per pulse. Driven at this current, Kr additions up to 1% show enhanced yields with a local maximum identified near 0.1% Kr. The yield-current scaling relationship with Kr dopants tends to be in excess of the sixth power, 6.45 ± 0.42 [dashed red line in Fig. 5(b)], based on only the two current levels studied here. This trend is interesting; however, it warrants more investigation at higher (and lower) current levels to elucidate an appropriate scaling relationship and further study what role Kr doping will play above the 3 MA-scale where yield is expected to drop precipitously.

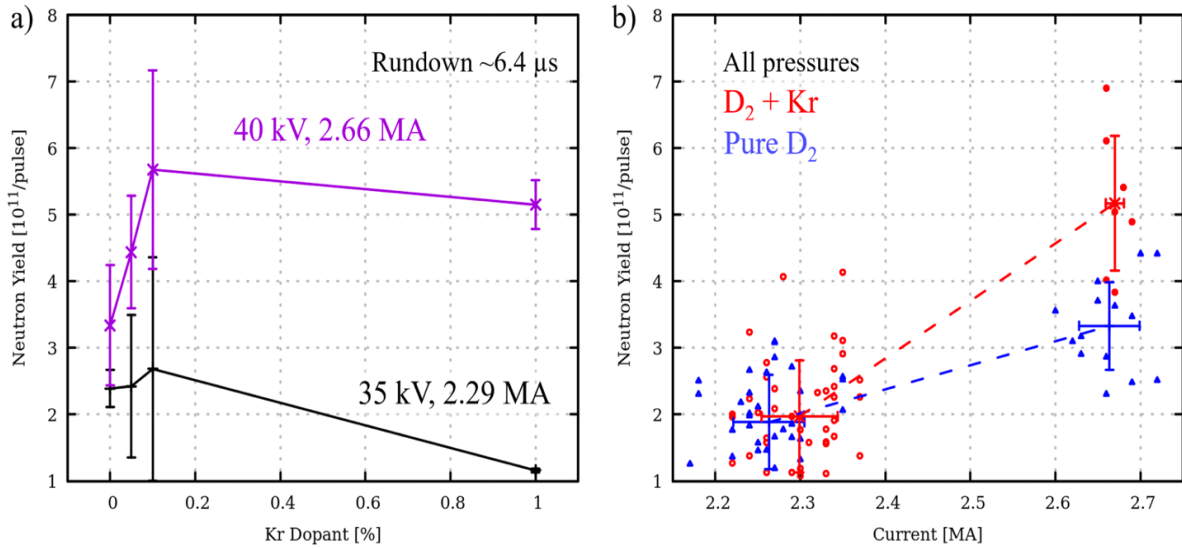


Figure 5: Current scaling of neutron production. (a) Variation in yield as a function of Kr dopant volume percentage. (b) Yield scaling as a function of current for both pure D_2 and Kr-doped admixtures.

Fig. 6 presents a short collection of results from the literature alongside the experiments performed under this grant for krypton doped deuterium DPF operations at the MA-scale. At 80 kA, an optimal krypton percentage for maximizing neutron yield was identified at $\sim 2\%$ by volume.²² For an optimized 95 kA machine, Talebitaher *et al.* [8] reported maximum neutron yields at $\sim 1\%$ Kr by volume. The experiments conducted by Aliaga-Rossel and Choi [9] (on a 780 kA DPF) only reported decreased yield; however, they only studied dopant percentages in excess of 0.5%, suggesting an optimal percentage could be between zero and 0.5% Kr. Combining the reported optimal percentages in the works cited above with our observed optimal percentage at $\sim 0.1\%$ at 2.29 MA and 2.66 MA, a trend emerges that suggests the optimal Kr dopant

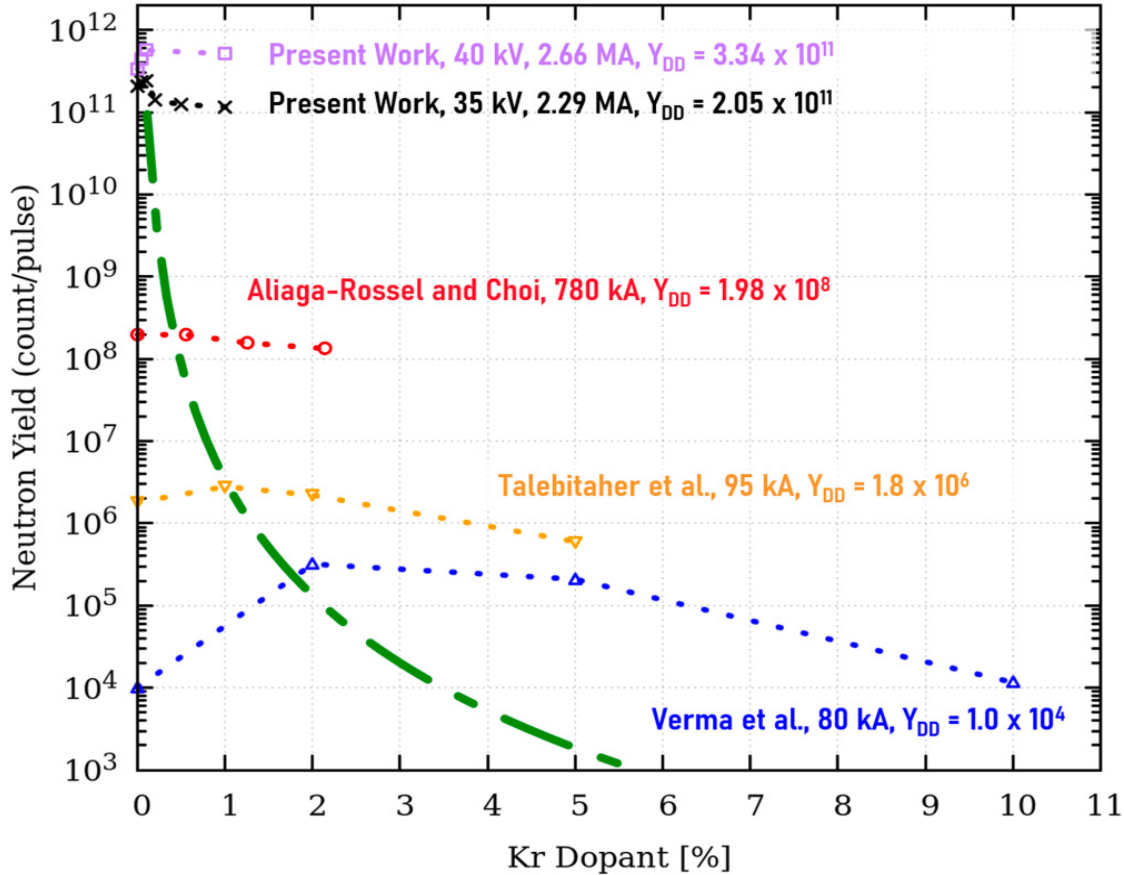


Figure 6: Neutron yield as a function of Kr dopant percentage for DPFs operated at five different current levels. The green line ($Y_N = 3.37 \times 10^6 \kappa^{-4.66}$) represents a first order fit of peak neutron production and corresponding.

percentage to be a function of total neutron yield (visualized in Fig. 6 by the dashed green line) and that the range of percentages over which Kr dopant is beneficial narrows significantly at higher yield and thus higher currents. The relationship between peak neutron yield (Y_N) and percent Kr volume fraction (κ) is well fitted by a first order approximation of an exponential function: $Y_N = 3.37 \times 10^6 \kappa^{-4.66}$.

Effect of insulator sleeve length on neutron yield

The length of the insulator sleeve plays a two-part role in the operation of a DPF. First, it affects the breakdown characteristics thereof, with several reports of optimal lengths and pressures that provide appropriate conditions for the formation of a uniform plasma sheath. Second, the insulator effectively shortens the length of the anode, reducing both the distance traveled and, thus, the time spent during the rundown phase. Changes to both time and distance will lead to a plasma sheath with varying mass and

velocity, which subsequently affects the final temperature and density (though not controlling them entirely). To better understand this careful balance and optimize neutron yield, we explored three insulator lengths at varying *deuterium fill pressure* using a 4.6 kJ DPF.

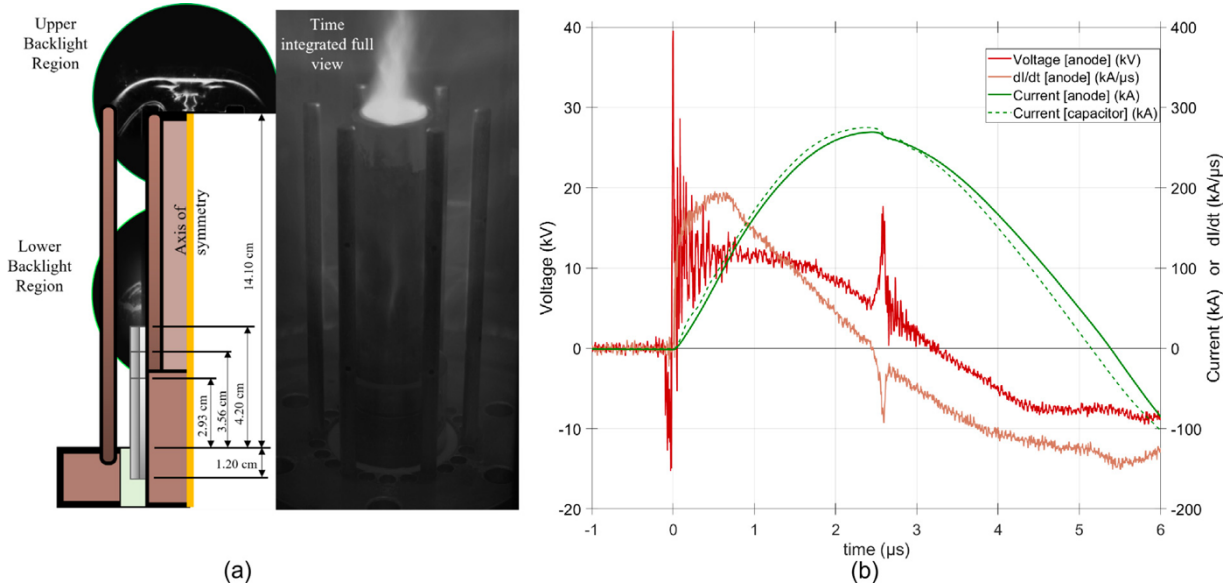


Figure 7: (a) Schematic drawing of the DPF electrode assembly. Each of the insulator lengths considered presently are illustrated alongside the upper and lower laser backlight probe regions. On the right-hand side, a time-integrated optical image of the entire region is represented. An additional time-integrated camera is focused on the pinch with additional filtering to resolve fine-scale features without saturation. (b) Typical current and voltage waveforms for the DPF experiment.

The experiments described herein were performed at UC San Diego on a newly constructed Mather-type DPF, which has characteristic circuit inductance $L \sim 113$ nH, capacitance $C \sim 19.2$ μ F, resistance $R \sim 6$ mOhm and is operated presently at a voltage $V \sim 22$ kV, resulting in 4.6 kJ of stored energy in the capacitor [see Fig. 7(a)]. The typical current rise time (quarter period) is ~ 2.460 μ s with a peak current on the order of 260 ± 13 kA [see Fig. 7(b)]. The total length of the central anode is 14.1 cm and is azimuthally surrounded by six cathode rods, 14.22 cm in length, spaced evenly around the axis. The anode radius is 1.55 cm with a 1.31 cm gap between the anode and the cathode rods. The insulator consists of two pieces, i.e., a Teflon adaptor and an interchangeable borosilicate insulator tube, which is the plasma-exposed insulator, henceforth referred to as the insulator sleeve. The insulator sleeve has an outer radius of 2.22 cm and an inner radius of 1.86 cm. The length of the insulator sleeve is varied from 2.93 to 4.20 cm to investigate changes in the insulator length on the performance of the DPF. These relevant dimensions and the position of optical measurements are graphically represented in Fig. 7. The DPF chamber vacuum is below 10^{-4} Torr before backfilling the chamber to between 1 and 10 Torr of deuterium gas. The system is triggered using a 15 kV, 100 ns rise time pulse connected to a pressure-controlled six-channel multi-gap (~ 13 mm total gap) switch that connects the capacitor to the central electrode.

Jitter in current start is ± 100 ns as measured from the arrival of a reference pulse from a delay generator and the current start. The voltage is measured using a voltage divider at the connection between the switch and the anode, and current is measured using two cross-calibrated Rogowski coils, one located below the switch and the second inside the chamber at the bottom of the anode. The data are digitally collected using a 16-channel 2-GS/s Tektronix TLS oscilloscope located in an adjacent control room within a Faraday cage.

To diagnose the plasma structure, schlieren imaging technique is used. For this purpose, two Quantel BigSky Nd:YAG lasers (10 ns FWHM, $\lambda = 532$ nm) are used to backlight 7.6 cm diameter circular projections of the DPF, referred to as the upper and lower backlights, respectively, with characteristic probe regions shown in Fig. 7. Both backlights are available for each shot, providing same-shot diagnostic capabilities spanning the early rundown to radial collapse phases. Each backlight is configured for schlieren imaging and digital images are collected using Canon EOS Rebel T6 single-lens reflex (SLR) cameras with a calculated spatial resolution of 25 μm per pixel. The upper backlight is setup as a two-frame system with a 50 ns delay leg providing a time evolution of the plasma pinch. The timing of the laser pulses is measured using silicon photodiodes. Additionally, SLR cameras are positioned to capture time-integrated images of visible emission produced by the plasma, the first of which provides an image of the entire

A reference measurement of neutrons was collected on a per-shot basis using a Be-activation detector designed by Mission Support and Test Services. The process is a two-step measurement of the activation of Be: incident neutrons cause alpha decay of the Be-9 into He-6, which in turn decays via beta emission into Li-6 with an 806 ms half-life. The beta particles (electrons) are collected by a photomultiplier tube biased at -2.2 kV with a 15 mV constant fraction discrimination threshold of the signal. Counts are collected via a multi-channel scalar processor, and the signals are binned in 160 ms intervals over 8 seconds to capture the expected decay as well as a measure of background noise (see Fig. 8). The primary components of the system were calibrated based on the continuous neutron source located at Sandia's Ion Beam Lab and the associated particle detectors available on site. In this configuration, each recorded half-life count corresponds to $\sim 1.1 \times 10^6$ neutrons over 4π .

Per-shot measurements provided the statistical foundation for optimization of the DPF in terms of DD fill pressure and with respect to the insulator sleeve length investigation, which will be presented in detail. The ability to optically diagnose the plasma was upgraded in year 3 by adding two independently-triggered Quantel BigSky Nd:YAG lasers (10 ns FWHM, $\lambda = 532$ nm) to backlight two 7.6 cm diameter circular projections of the DPF. Both backlights are available for each shot and are configured for schlieren imaging,

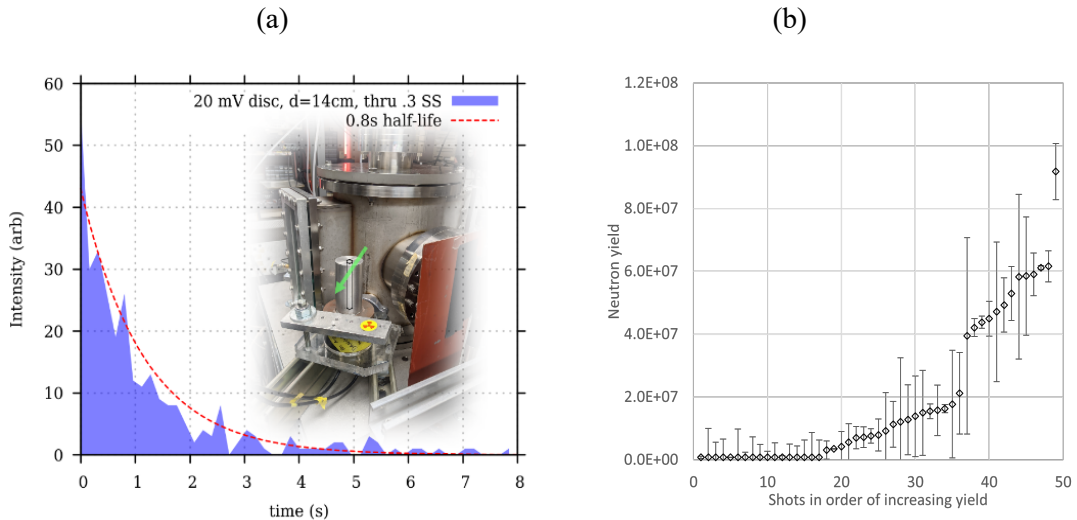


Figure 8: (a) Representative measurement of Be-activation detector measurement. The inset illustrates location and orientation of the Be-rods relative to the pinch. The green arrow represents the direction of the neutrons hitting the detector. (b) Shots using the standard insulator for filling pressures between 4.5 and 5.5 torr, in order of increasing neutron yield. One in three shots lies below the lower detection limit. The average and error bars are based on the confidence interval of an exponential decay fit for the counts at $t=0$.

providing same-shot diagnostic capabilities spanning the early rundown to radial collapse phases. Two additional SLR cameras were positioned to capture time-integrated images of visible emission produced by the plasma, the first of which provides an image of the entire assembly, while the second is focused on the pinch region with additional neutral density filters to resolve finer scale plasma structures without saturation.

A comparison of the neutron yield as a function of insulator length and fill pressure is shown in Fig. 9, which will be used to further optimize the neutron yield of the system in future studies.

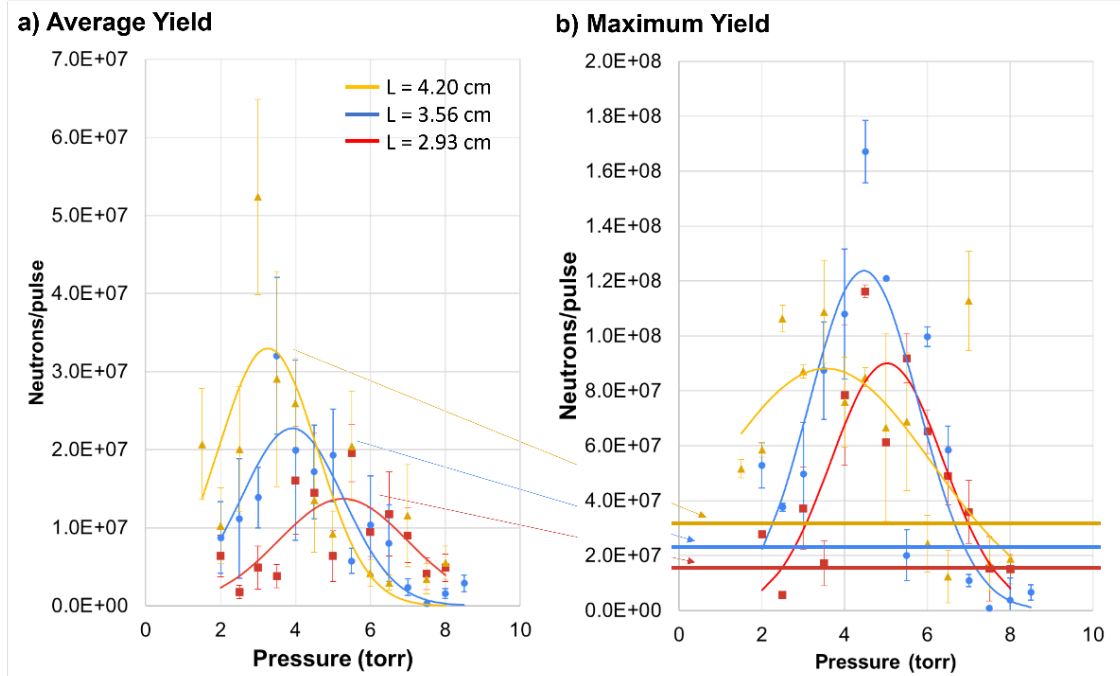


Figure 9: Average (a) and maximum (b) neutron yield produced per shot as a function of deuterium fill pressure for insulator lengths of 2.93, 3.56, and 4.20 cm. Solid lines are Gaussian fits to the data. The highest average yields at each insulator length are represented as horizontal bars on the maximum yield plot (y-scale reference).

The single highest yielding shot was from the intermediate insulator sleeve at 4.5 torr, reaching $1.7 \pm 0.1 \times 10^8$ neutrons, with the maximum neutron yields for each insulator length on the order of 10^8 neutrons/shot. This value is approximately five times the average and greater than three times the standard deviation. If the yield distribution for each configuration was Gaussian, the probability of obtaining one such shot would be $<0.3\%$, while just 10-15 shots were taken for each configuration. It is expected that “optimized” DPFs have comparable drive parameters, $I_p/(a\sqrt{\rho_0})$, where I_p is the peak current, a is the anode radius, and ρ_0 is the fill gas pressure. The average drive parameter of optimized machines (typically ~ 4.75 torr) is $90 \text{ kA/cm} \cdot \text{torr}^{1/2}$ (varying between 75 and $105 \text{ kA/cm} \cdot \text{torr}^{1/2}$) and an average axial velocity (v_a) of 70 km/s, calculated by $v_a = L \cdot I_p / (a\sqrt{\rho_0})$. In the present configuration, the DPF experiments reported on herein have a drive parameter of $75\text{-}82 \text{ kA/cm} \cdot \text{torr}^{1/2}$ at the 4.5-5.5 torr fill pressure range. Between ~ 2.8 and 5.4 torr, the DPF configuration lies within the optimal drive parameter range for neutron production.

Critical drive properties were evaluated on a per shot basis to assess the conditions necessary for a successful pinch. A significant change in pinch time was observed, which is directly linked to the length of

the insulator length for a fixed pressure. This demonstrates that the change in insulator length can be

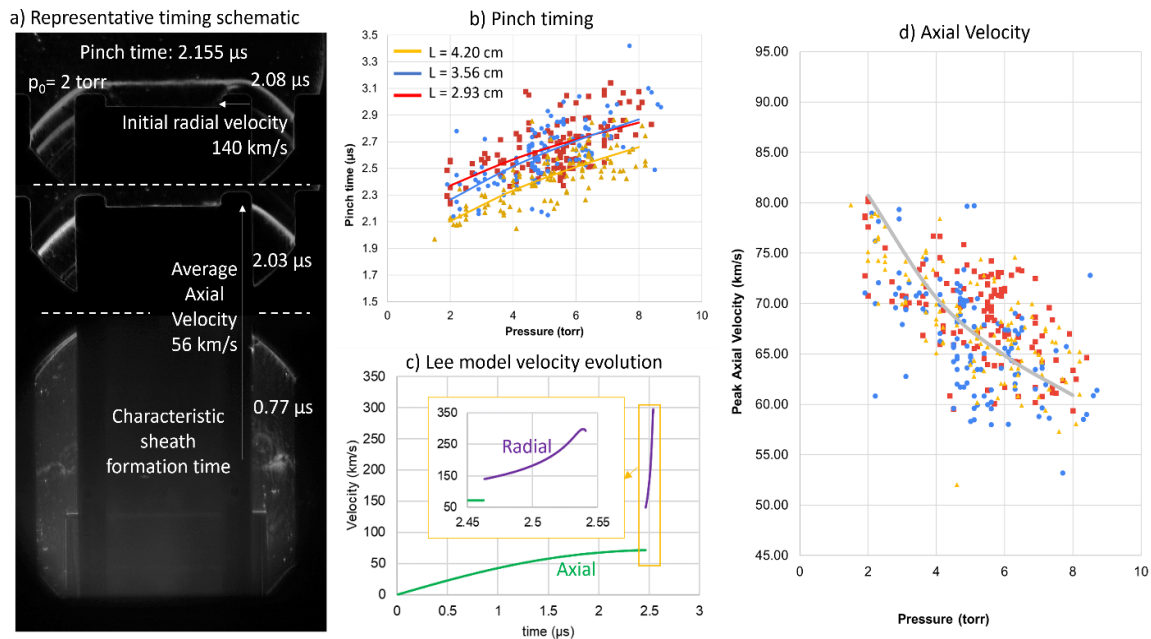


Figure 10: a) Representative timing schematic showing evolution of current sheath from phase to phase; times listed are evaluated from a single shot. b) Experimentally measured pinch times as a function of fill pressure for each insulator length. Curved lines are fit according to the square root of pressure. c) Lee model calculated time-dependent velocity. d) Peak axial velocity with Lee model prediction shown in gray.

correlated to a reduction in anode length. Worth of note is the shot-to-shot variation in axial velocity (and thus radial velocity), and the consistency among the three insulator lengths. Expectedly, there was an observed change in sheath velocity with fill pressure (see Fig. 10).

Critically, accurate experimental data fitting required a variable mass sweeping factor (massf equivalent to $0.4 \cdot p_0^{-1/2}$ for all insulators, akin to the relationship expected between sheath velocity and pressure, although massf is typically a static fitting variable. This is the first instance where this correction was required, according to literature. This suggests that considerably less mass is effectively swept at higher fill pressures, for instance massf equal to 0.28, 0.2, and 0.16 for 2, 4, and 6 torr respectively, with significant ramifications of future DPF strategies that strive to increase neutron production by increasing the target density via high DD pressure operation.

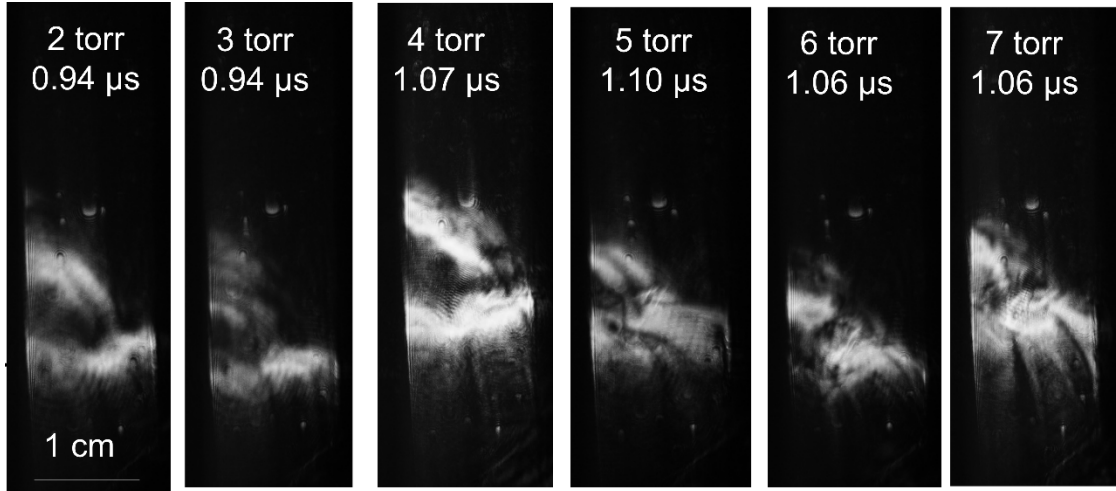


Figure 11: Structure of initial plasma sheath at the beginning of the axial rundown phase ($\sim 1 \mu s$). From left to right the effect of fill pressure can be observed, i.e. increasing filamentation effects with increasing pressure.

Another critical observation was the direct identification of finger-like striations that become more pronounced with increasing fill pressure, as shown in Fig. 11. These features are characteristic of plasma filamentation, which may be a manifestation of partial current sheath ionization. This was detailed by Gratton *et al.* and Milanese [10, 11] to explain a reduction in neutron yield at higher fill pressures, but to date is poorly understood. As an indirect measure of this effect, the experimental data was fit by the Lee model to provide a time-dependent relationship of the electromechanical work (E_p), analytically formulated as:

$$E_p = \frac{1}{2} \int \dot{L} I^2 dt$$

where $\dot{L}$ is the rate of change of inductance, I is the current, and the integral thereof multiplied by one half gives the electro-mechanical work applied to the plasma. As a numerical solution, we integrate for each timestep calculated using the optimized Lee model with the critical relationship between fill pressure and massf defined above. The connection between fill pressure and breakdown is often understood from a Paschen-law relationship, wherein three operating pressure regimes are commonly defined: low-pressure ($p_0 < 0.75$ torr) defining diffuse/uniform volume discharge; medium pressure ($0.75 < p < 3.75$ torr) favoring insulator surface e^- glide/avalanche; and high pressure ($p_0 > 3.75$ torr) producing radial filaments that bridge the anode and cathode. However, the connection between Paschen-type physics and DPFs is fragile - the Paschen curve refers to a specific situation where ions in a gas produced during ionization travel to the cathode and release sufficient secondaries to cause a feed-back loop. Notably, this is not the type of breakdown that occurs in DPFs and thus, once a plasma is formed, such physics should no longer apply. In general, these pressure boundaries are variable guidelines proportional to the geometry of a specific DPF and likely empirically consider surface physics effects.

As a final consideration, the breakdown physics will depend on the evolving temperature of the initial plasma (typically a few eV) which define the mean free electron paths. Assuming a $T_e \sim 4$ eV for all fill pressures, and using the relationships between ionization path length and pressure, $L_{iz}(P)$, derived by

Bennett *et al.*[12], it is found to agree remarkably well with $L_{iz}(P)/L_i = 2.4$ when using the optimal pressure for each of the three insulator lengths considered in our study. These ionization path lengths are larger than expected for optimal breakdown based on a neutral fill gas – indicative of a significant contribution of the insulator material surface effects. The fixed ratio of L_{iz} to L_i is supportive of the concept that the optimal insulator length for breakdown/lift-off may scale with the ionization path length and provides a direct means to optimize DPF operations moving forward as geometries are changed.

There exists an optimal intermediate pressure regime wherein neutron production is increased, the lower boundary of this pressure determined predominantly by the drive parameter of the system. An upper bound on pressure is identified whereafter significant filamentation is observed as related to the partial ionization of the system during initial sheath formation and liftoff. Increasing insulator length requires lower fill pressures to optimize the average neutron yield, in contrast to what is expected based on maintaining pinch timing. These factors in concert represent the careful balance between the electrode-insulator geometries and the operating pressure range. Future studies will evaluate specific shot-to-shot variations observed and changes to the anode-cathode geometry as they relate to efforts towards the further optimization of neutron generation.

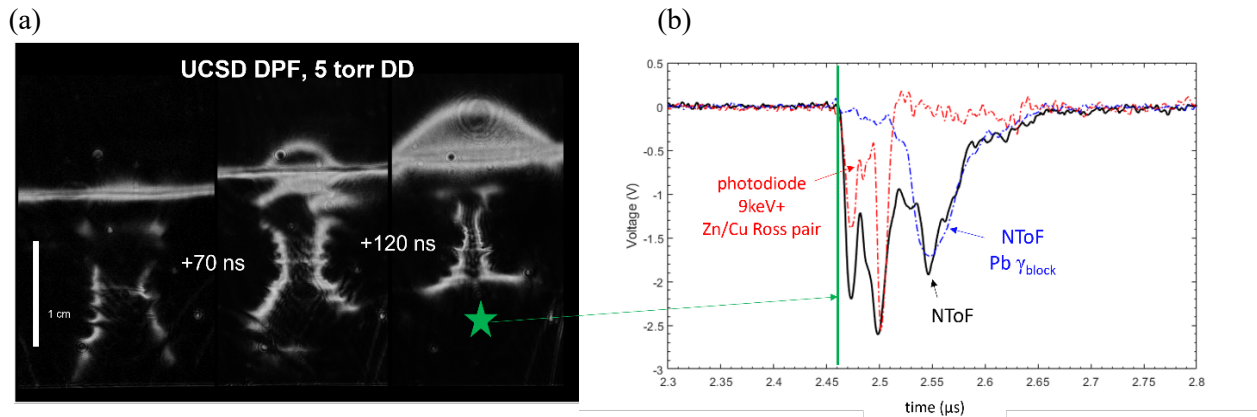


Figure 12: (a) A series of schlieren images of pinch at the top of anode, (b) Time of flight neutron signals with and without lead filter.

A complementary set of time-resolved neutron detectors were also implemented to measure the emission properties of the DPF, with the ability to separate the gamma and neutron signals that are typically observed during DPF operation using neutron time-of-flight (NToF) diagnostics. An example of NToF signals is shown in Fig. 12(b) along with the X-ray signal recorded by a filtered photodiode. As a reference, schlieren images are shown in Fig. 12(a). The time of the last image was coincident with the start of X-ray production, which is recorded both by the photodiode and an unshielded NToF detector. The X-ray signal is comprised of two peaks at 2.47 and 2.50 μs. A third peak at 2.55 μs corresponds to the detection of neutrons. A second NToF detector, placed behind a Pb shield, only recorded the neutron signal indicating that the X-rays were successfully stopped by the shield.

Role of anode geometries on pinch dynamics and neutron generation

Experiments to investigate three different anode geometries were carried out with a 4.6 kJ, 20 kV, 230 kA DPF. Solid and hollow anodes with either a flat or hemispherical tip were explored. A series of time-gated laser schlieren images, filtered X-ray photodiodes, and neutron activation detectors were used. We show

first evidence of the extension of the pinch column inside the hollow anode, presumably responsible for higher yield. A maximum neutron yield of $(2.0 \pm 0.2) \times 10^8$ neutrons per pulse was observed with the hollow hemispherical anode, while higher hard X-ray emissions were observed for the solid hemispherical anode. This analysis leads to the conclusion that a hollow anode optimizes neutron yield while a solid anode serves as an enhanced X-ray source due to electron beams.

The three anodes in this experiment are shown in Fig. 13, which were fabricated out of copper; each anode had a length of 10.8 cm, an outer diameter of 3.15 cm, and for those anodes with a hollow, and inner diameter of 1.52 cm. To our knowledge, this work premieres the use of a flat end hollow anode with slits which was designed to provide a line of sight into the anode and capture potential plasma activity within the hollow. Experiments for each anode were run over a pressure scan from 3.0 to 8.0 torr in increments of 0.5 torr.

In addition to neutron and laser probing diagnostics discussed above, an X-ray PIN photodiode of an area of 1 mm^2 and a $10 \text{ }\mu\text{m}$ aluminum filter was placed in the radial direction of the anode tip and was used to capture X-rays signal.

Over two hundred shots with each anode were conducted to gather neutron and X-ray data as well as time-integrated and schlieren images at each pressure. On average, the hemispherical hollow anode produced more neutrons compared to the other anodes, with maximum yield of $(2.0 \pm 0.2) \times 10^8$ at 6.5 torr. The hemispherical solid anode had a maximum neutron yield of $(1.1 \pm 0.1) \times 10^8$ at 5 torr but overall, averaged lower neutron yields than the other two anodes. The maximum and average neutron yields for each of the anodes for the range of pressures from 3.0 torr to 8.0 torr are shown in Fig. 14.

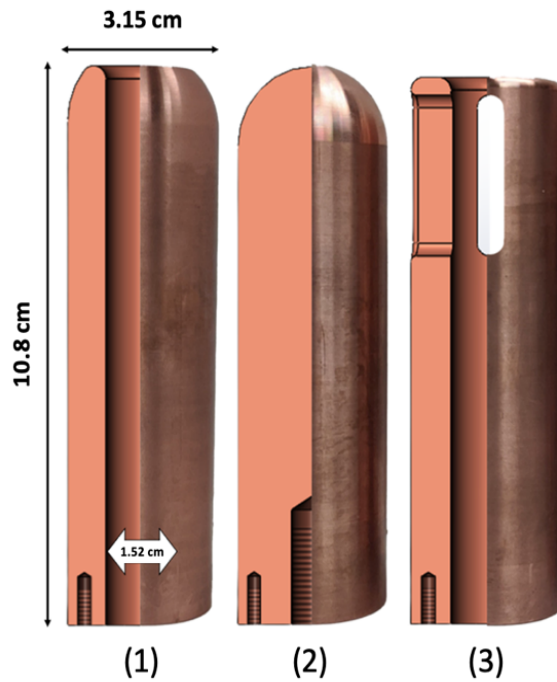


Figure 13: Anodes from left to right: 1) hemispherical hollow anode, 2) hemispherical solid anode, and 3) flat end hollow anode with slits. The figure shows each anode represented by its camera image (right half) and its computer-aided design (CAD) image (left half).

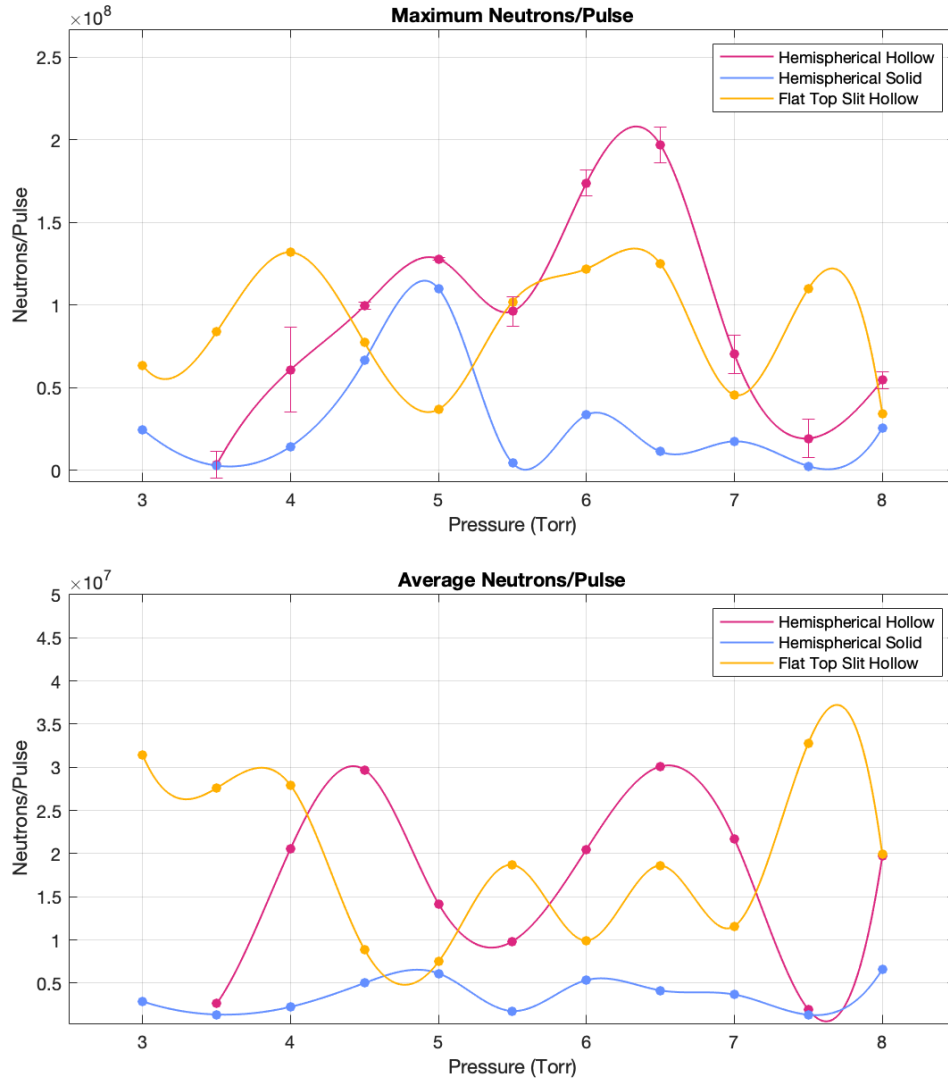


Figure 14: Maximum neutron yields (top) and average neutron yields (bottom) for the three anodes over a pressure range of 3.0 to 8.0 torr at 0.5 torr increment.

Care was taken to exclude from averages the shots that may have had external effects influencing neutron yield. For instance, the first shots of the day and after any atmospheric venting of the vacuum chamber were meant to re-condition the electrodes and flush out any impurities that may have been absorbed. These shots were therefore discarded when calculating the neutron averages. A variety of trends can be seen in the figures above including a dip in neutron yield over the pressure range of 5.0 torr to 5.5 torr. More detailed studies would be required to fully characterize this phenomenon. The operating fill pressure is also an important parameter with a significant influence on neutrons and the plasma sheath structure²¹. Filamentary structures identified in the plasma sheath is seen to become more prominent as the fill pressure increased.

X-ray signals were captured using a PIN photodiode located radially from the tip of the anodes that used a 10 μm aluminum filter. Fig. 15 shows the maximum voltage peaks captured for both the hemispherical hollow and solid anodes over a pressure range of 3.0 torr to 8.0 torr and Fig. 16 shows the measured maximum and time integral of the photodiode signal captured for both anodes.

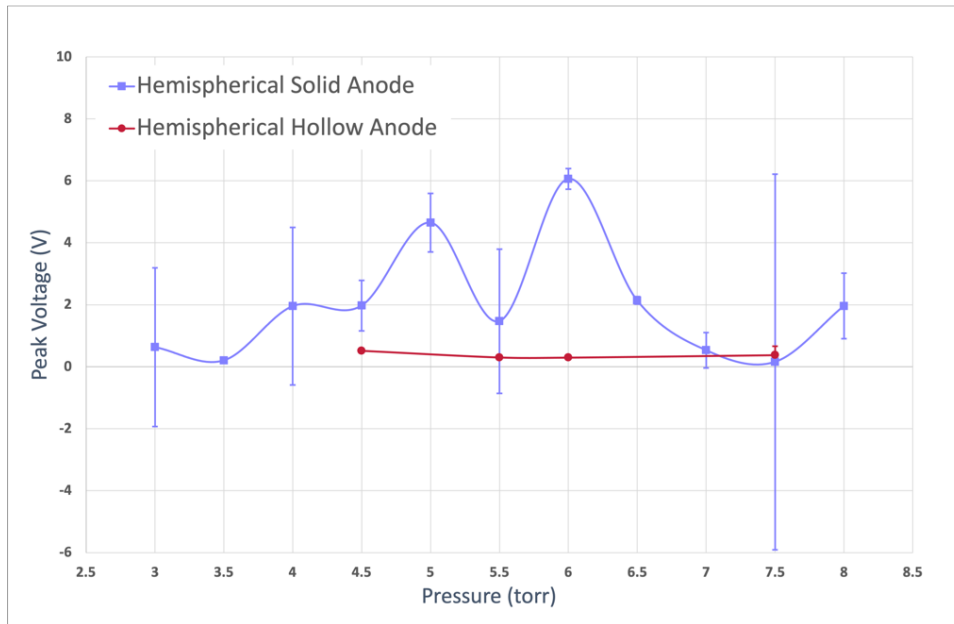


Figure 16: Pressure (torr) versus peak voltage signal (volts) captured by a PIN photodiode with an area of 1 mm^2 and a $10 \text{ }\mu\text{m}$ aluminum filter for both hemispherical anodes. The diode was placed in the radial direction from the tips of the anodes. The data is taken at a pressure range from 3.0 to 8.0 Torr; however, the hemispherical hollow anode did not capture diode signals at each point of this range.

The distinct contrast in diode signals and neutron yields from the two anodes demonstrates how the presence and absence of a hollow structure in the central anode has a strong influence on energetic particle and X-ray yields. The hemispherical solid anode had significantly higher X-ray emission and lower neutron yield compared to the hollow anode. The higher X-ray emission is likely due to the presence of copper target at the end of anode, interacting with an electron beam. This is consistent with B.H. Shaw et al. [13] work where significantly higher copper sputter was observed in case of solid anode compared to hollow anode. In fact, they measured sputter mass of $16.9 \pm 1.0 \text{ ng per sr per discharge}$ of copper compared to $117 \pm 64.8 \text{ }\mu\text{g per sr per discharge}$ when additional mass was added. In our case, the weaker pinches directly relate to copper impurities in the pinch consistent with H.B. Shaw *et al.* [13] observation where they recorded drop in neutron yield with the additional mass provided by a solid anode in relation to a hollowed anode. They conjectured that lower neutron production may be due to radiation loss reducing the plasma temperature and hence D-D fusion cross-section. Our observations agree with this conjecture. Figure 17 further quantifies the contrast in X-ray emissions between the hemispherical solid and hemispherical hollow anodes as it shows the calculate area under each voltage spike which is directly related to flux due to electrons.

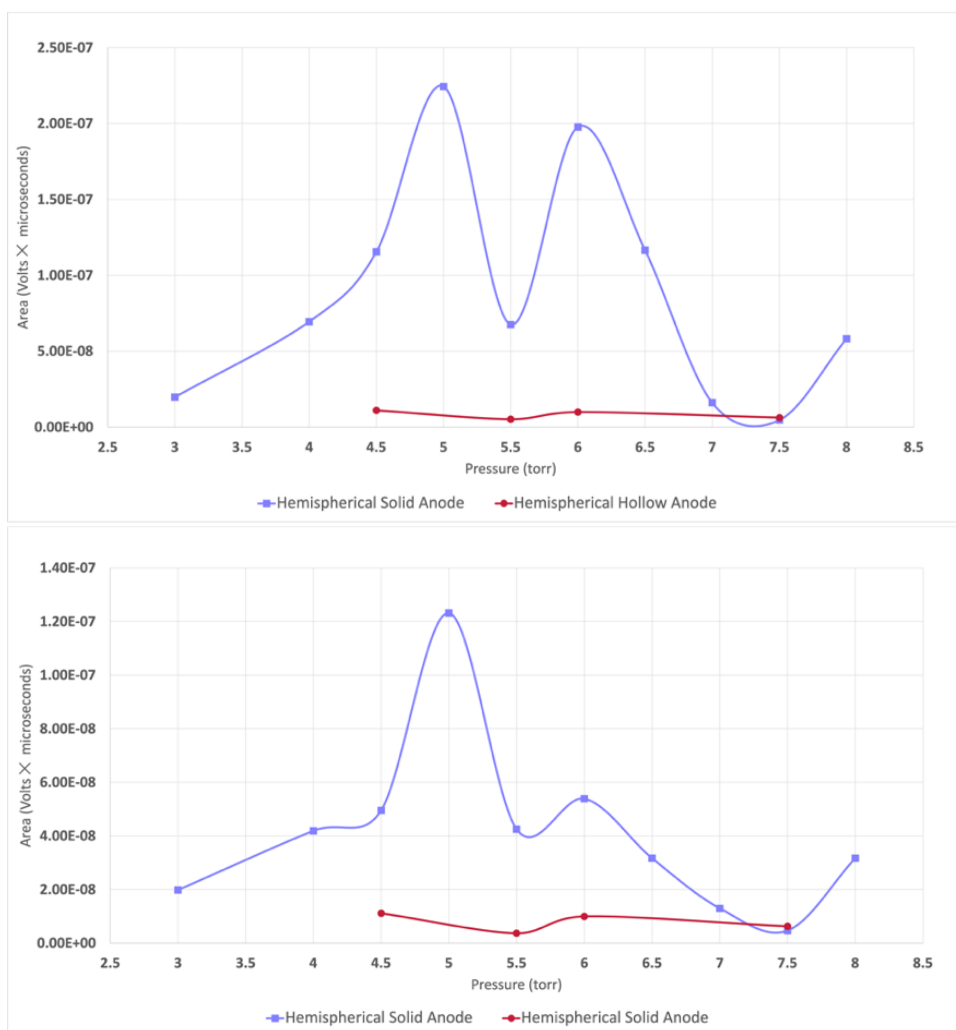


Figure 17: The total flux captured is measured by calculating the time integral of the photodiode signal. The maximum area under voltage spike (top) and average area under voltage spike (bottom) for the two hemispherical anodes over a pressure range of 3.0 to 8.0 torr are presented.

Laser schlieren and optical time-integrated images were also analyzed in order to visually characterize pinch structures for each anode shape. Fig. 18 shows visible light emission from time-integrated images of the pinch. The hemispherical solid shows a significantly brighter and asymmetrical pinch column whereas the hollow anodes tend to be fainter and axially aligned

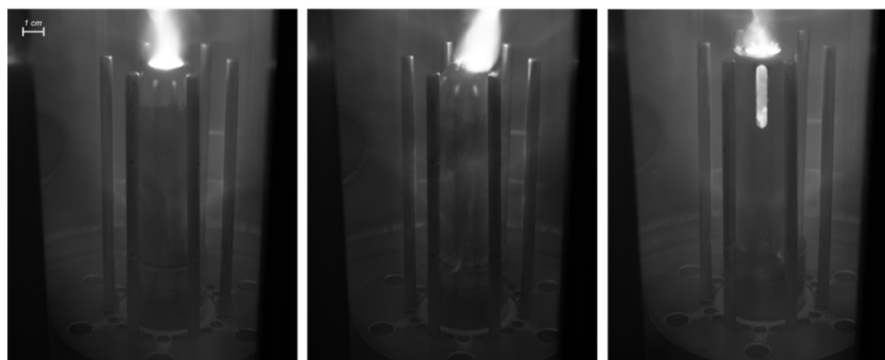


Figure 18: Time-integrated images of each of the anodes showing bright visible light emitted during implosion at a pressure of 5.0 torr. The emission caused by the hemispherical solid anode is brighter and asymmetric compared to the other two anodes. Light is seen coming out of the front opening of the flat end hollow anode with slits, indicating that plasma activity is present within the anode.

with the anode.

Schlieren images facilitate useful information about the pinch dynamics and may provide insight on neutron production. If a dense pinching column is seen at the tip of the anode, it implies that a significant number of neutrons may have been produced. Fig. 19 shows the development of a pinch for each of the three anodes through a series of schlieren images of distinct shots taken with the same fill pressure. The pinch dynamics of the sheath during the “Axial Rundown” phase are similar for all three anodes. The critical distinction that leads to the difference in neutron yield comes during the “Radial Implosion” phase. The pinches formed with the hollow anodes were symmetric and dense whereas the solid anode does not successfully form a pinch. The plasma sheath that forms with the solid hemispherical anode does not compress and form a pinch, instead, bright visible light is seen expelling from the tip of the anode.

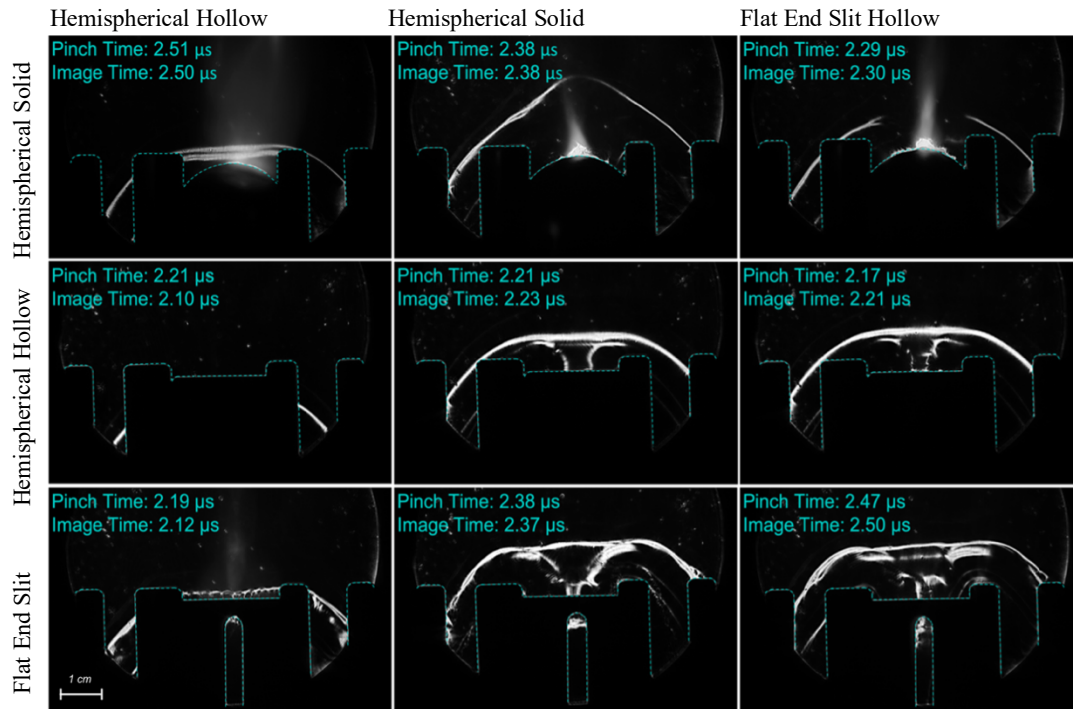


Figure 19: Three rows of schlieren images showing the evolution of a pinch for the given anode labeled on the left. Each image provides the time in which a voltage spike is detected (pinch time) and the time in which the image was captured (image time) for the corresponding shot.

The flat end hollow anode with slits offers a new perspective of the plasma activity occurring within the anode that has not been seen in previous works. When examining the schlieren images for this anode, it is evident that the plasma sheath is visible through one of the slits moving in the axial direction before reaching the radial implosion phase. The extended pinch in hollow anode allows the pinching process continuation inside the anode. The hemispherical solid anode lacks the necessary space for an elongated pinch to form and the dome-tip geometry tends to obstruct the sheath from forming a structured pinch. The reduction in radius due to the anode’s hemispherical tip geometry is the reason behind the shortening of the plasma column which makes the solid hemispherical anode unsuitable to produce significant neutron yields. We believe this is one of the reasons for the reduction in neutron yield. A comprehensive MHD simulation will help to answer some of these questions.

4. Changes

All the tasks defined in the proposal were met except magnetic field measurement using Faraday rotation. New scaling was discovered with dopant gases, which needed additional computational work to validate experimental data. We plan to include measurement of magnetic field in the new proposal.

5. Publications

- [1] E. N. Hahn, D. Housley, J. Narkis, F. Conti, D. R. Lowe, and F. N. Beg, “Effect of krypton admixture in deuterium on neutron yield in a megaampere dense plasma focus,” *J. Appl. Phys.***128**, 143302 (2020); doi: 10.1063/5.0020936.
- [2] J. Narkis, E. N. Hahn, D. R. Lowe, D. Housley, F. Conti, and F. N. Beg, “Magnetohydrodynamic simulations of a megaampere-class Kr-doped deuterium dense plasma focus,” *Phys. Plasmas* **28**, 022707 (2021); doi: 10.1063/5.0033129.
- [3] D. Housley, E. N. Hahn, J. Narkis, J. R. Angus, A. J. Link, F. Conti, F. N. Beg, “Effect of insulator surface conditioning on the pinch dynamics and x-ray production of a Ne-filled dense plasma focus,” *J. Appl. Phys.***129**, 223303 (2021); doi: 10.1063/5.0050203.
- [4] E. N. Hahn, S. Ghosh, V. Eudave, J. R. Angus, A. J. Link, F. Conti, F. N. Beg, “Effect of Insulator Length and Fill Pressure on Filamentation and Neutron Production in a 4.6 kJ Dense Plasma Focus,” *Phys. Plasmas* **29**, 083508 (2022); doi: 10.1063/5.0087901.
- [5] V. Eudave, F. Conti, S. Ghosh, M. P. Valdivia, E. N. Hahn, Justin Angus, J. Vaughan, and F. N. Beg, “Effects of solid and hollow anode shapes on neutron yield in dense plasma focus,” *IEEE Transaction on Plasma Science* (under review).

6. References

- [1] Y. Kato and S. H. Be, *Appl. Phys. Lett.***48**, 686 (1986).
- [2] M. Krishnan, *IEEE Trans. Plasma Sci.***40**, 3189–3221 (2012).
- [3] N. V. Filippov, T. I. Filippova, and P. V. Vinogradov, *Nucl. Fusion* **2**, 577 (1962).
- [4] J. W. Mather, *Phys. Fluids* **7**, S28 (1964).
- [5] A. Bernard, P. Cloth, H. Conrads, A. Coudeville, G. Goullan, A. Jolas, C. Maisonnier, and J. P. Rager, *Nucl. Instruments Methods* **145**, 191–218 (1977).
- [6] F. Hohl and S. P. Gary, *Phys. Fluids* **20**, 683 (1977).
- [7] W. Stygar, G. Gerdin, F. Venneri, and J. Mandrekas, *Nucl. Fusion* **22**, 1161–1172 (2011).
- [8] A. Talebitaher, S. Lee, S. M. P. Kalaiselvi, R. Verma, P. Lee, S. V. Springham, T. L. Tan, and R. S. Rawat, *J. Fusion Energy* **35**, 370 (2016).
- [9] R. Aliaga-Rossel and P. Choi, *IEEE Trans. Plasma Sci.* **26**, 1138 (1998).
- [10] R. Gratton, H. Kelly, M. Milanese, and J. Pouzo, *Phys. Lett. A* **62**, 422 (1977).
- [11] M. M. Milanese, J. J. Niedbalski, and R. L. Moroso, *IEEE Trans. Plasma Sci.* **35**, 808 (2007).
- [12] Bennett, M. Blasco, K. Breeding, V. DiPuccio, B. Gall, M. Garcia, S. Gardner, J. Gatling, E. C. Hagen, A. Luttmann, B. T. Meehan, S. Molnar, R. O’Brien, E. Ormond, L. Robbins, M. Savage, N. Sipe, and D. R. Welch, *Phys. Plasmas* **24**, 062705 (2017).
- [13] B.H. Shaw, S. Chapman, C.M. Cooper, C. Goyon, J. Angus, A. Link, D.P. Higginson, J.X. Liu, J.M. Mitrani, Y.A. Podpaly, A. Povilus, and A. Schmidt, *Journal of Applied Physics* **124**, 233301 (2018).